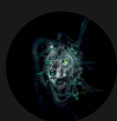


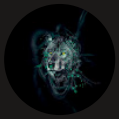


Load Data into a DynamoDB Table



Kiprono Yegon

☐	<i>Id (Number)</i>	<i>Authors</i>	<i>ContentType</i>	<i>Difficulty</i>	<i>Price</i>	<i>ProjectCategory</i>	<i>Published</i>	<i>Services</i>	<i>Title</i>	<i>URL</i>	<i>VideoType</i>
☐	3	[{"S": "Ne...	Project	Easy peasy	0	AI/ML	true		Build a Cha...	aws-ai-lex1	
☐	2	[{"S": "Ne...	Project	Easy peasy	0	Analytics	true		Visualize da...	aws-analyti...	
☐	203		Video		0			[{"S": "Am...	AWS x Data...	https://you...	Live Project Demo
☐	202		Video		0				Don't miss ...	https://you...	Shorts
☐	201		Video		0			[{"S": "Am...	AWS Relati...	https://you...	Educational
☐	1	[{"S": "Nat...	Project	Easy peasy	0	Storage	true		Host a Web...	aws-host-e...	



Introducing Today's Project!

What is Amazon DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database service that stores and retrieves large amounts of data quickly and efficiently it is important because of its speed and flexibility

How I used Amazon DynamoDB in this project

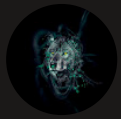
In today's project, I used Amazon DynamoDB to create tables and load data into the attributes in the tables

One thing I didn't expect in this project was...

One thing I didn't expect in this project is the Ease of loading data to the tables

This project took me...

This project took me about thirty minutes



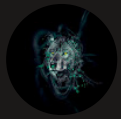
Create a DynamoDB table

DynamoDB tables organises data using a list, each item with their own list of attributes

An attribute is a piece of data about an item and can have multiple attributes.

The screenshot displays the AWS DynamoDB console interface for the 'NextWorkStudents' table. On the left, a sidebar shows 'Tables (1)' with 'NextWorkStudents' selected. The main panel is titled 'NextWorkStudents' and features a 'Scan or query items' section. In this section, 'Scan' is selected, and the 'Table' dropdown is set to 'Table - NextWorkStudents'. Below this, a green status bar indicates 'Completed' with 0 items returned and 0 items scanned. A table titled 'Table: NextWorkStudents - Items returned (1)' shows a single item with 'StudentName (String)' as 'Nikko' and 'ProjectsComplete' as 4. The scan started on May 16, 2026, at 21:57:10.

StudentName (String)	ProjectsComplete
Nikko	4



Read and Write Capacity

Read capacity units (RCUs) gives a way to measure how many engines DynamoDB is using to operate. and write capacity units (WCUs) give the DynamoDB tables the engines to edit/update/delete data!

Amazon DynamoDB's Free Tier gives you 25GB of data storage, plus 25 Write and 25 Read Capacity Units (WCU, RCU). This is enough to handle 200M requests per month I turned off auto scaling because it has the power to boost your table's processing power increasing the costs

Read/write capacity settings [Info](#)

Capacity mode

☐ On-demand
Simplify billing by paying for the actual reads and writes your application performs.

☒ Provisioned
Manage and optimize your costs by allocating read/write capacity in advance.

Read capacity

Auto scaling [Info](#)
Dynamically adjusts provisioned throughput capacity on your behalf in response to actual traffic patterns.

☐ On
☒ Off

Provisioned capacity units

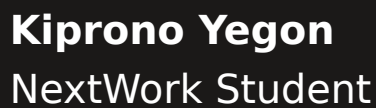
5

Write capacity

Auto scaling [Info](#)
Dynamically adjusts provisioned throughput capacity on your behalf in response to actual traffic patterns.

☒ On
☐ Off

Minimum capacity units	Maximum capacity units	Target utilization (%)
1	10	70



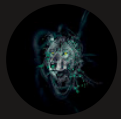
Using CLI and CloudShell

AWS CloudShell is shell in your AWS Management Console, which comes with pre-installed AWS CLI

AWS CLI is a software that lets you create, delete and update AWS resources with commands instead of clicking through your console.

I ran a CLI command in AWS CloudShell that created four tables, each with specific attributes and settings.

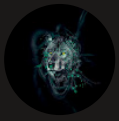
[illegible]



Loading Data with CLI

I ran a CLI command in AWS CloudShell that loads multiple items into DynamoDB tables

```
    ]
  }
}
nextworksampladata $ aws dynamodb batch-write-item --request-items file:///ContentCatalog.json
{
  "UnprocessedItems": {}
}
nextworksampladata $
nextworksampladata $ aws dynamodb batch-write-item --request-items file:///Forum.json
{
  "UnprocessedItems": {}
}
nextworksampladata $
nextworksampladata $ aws dynamodb batch-write-item --request-items file:///Post.json
{
  "UnprocessedItems": {}
}
nextworksampladata $
nextworksampladata $ aws dynamodb batch-write-item --request-items file:///Comment.json
{
  "UnprocessedItems": {}
}
nextworksampladata $ █
```



Observing Item Attributes

Attributes

Add new attribute ▼

Attribute name	Value	Type	Remove
Id - Partition key	1	Number	
Authors	Insert a field ▼	List	Remove
ContentType	Project	String	Remove
Difficulty	Easy peasy	String	Remove
Price	0	Number	Remove
ProjectCategory	Storage	String	Remove
Published	☒ True ☐ False	Boolean	Remove
Title	Host a Website on Amazon S3	String	Remove
URL	aws-host-a-website-on-s3	String	Remove

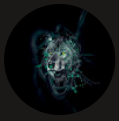
Cancel

Save

Save and close

I checked a ContentCatalog item, which had the following attributes:Authors, ContentType, Difficulty, Price, ProjectCategory, Published, Title and URL

I checked another ContentCatalog item, which had a different set of attributes from the first table, it has no StudentsComplete



Benefits of DynamoDB

A benefit of DynamoDB over relational databases is flexibility, because every item having their own unique set of attributes is a huge advantage when items in a table could look different from each other.

Another benefit over relational databases is speed, because they use partition keys to split up a table and quickly find the items they're looking for.

☐	Id (Number)	Authors	ContentType	Difficulty	Price	ProjectCategory	Published	Services	Title	URL	VideoType
☐	3	[{"S": "Ne...	Project	Easy peasy	0	AI/ML	true		Build a Cha...	aws-ai-lex1	
☐	2	[{"S": "Ne...	Project	Easy peasy	0	Analytics	true		Visualize da...	aws-analyti...	
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☐	1	[{"S": "Nat...	Project	Easy peasy	0	Storage	true		Host a Web...	aws-host-e-...	



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